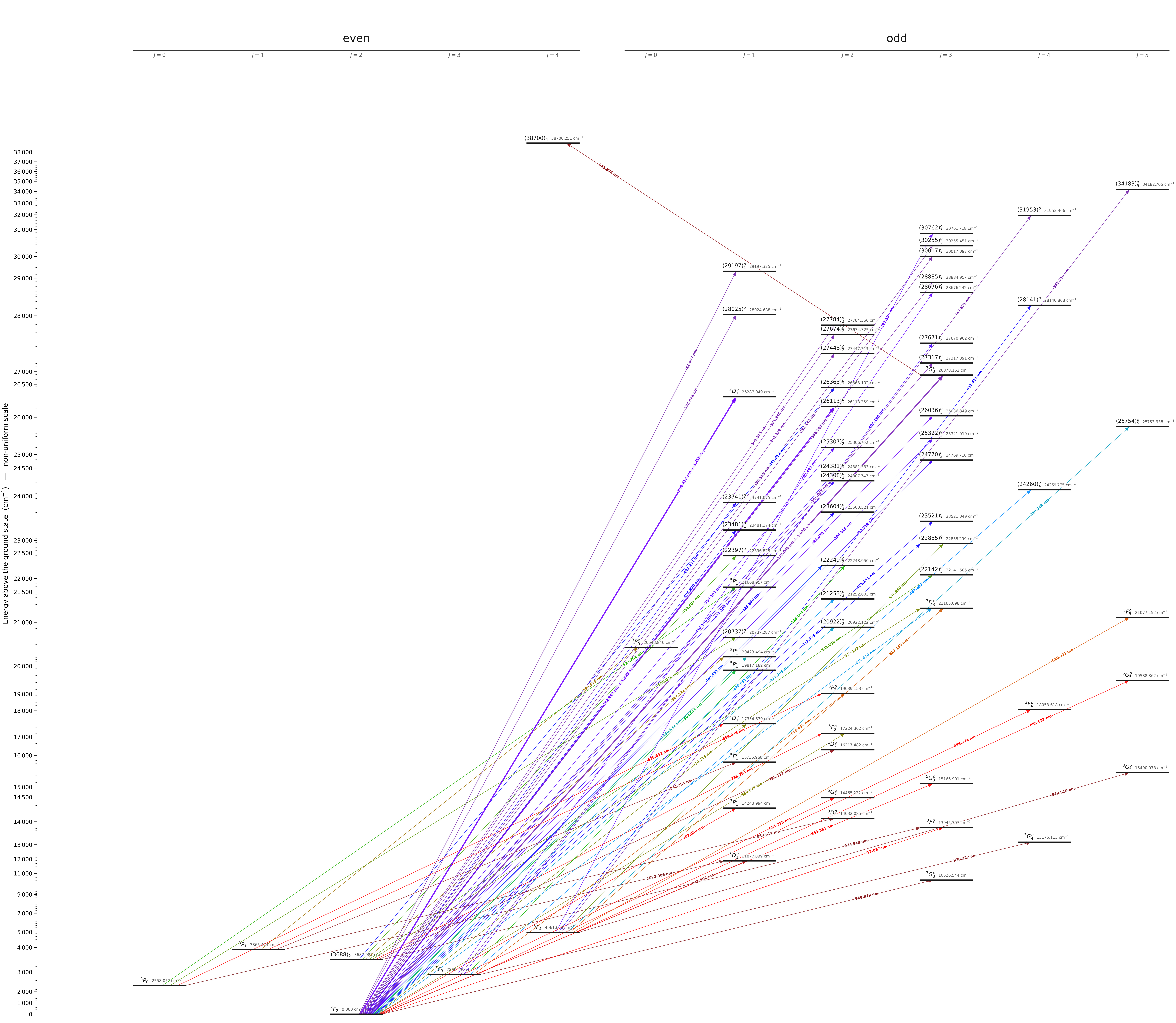
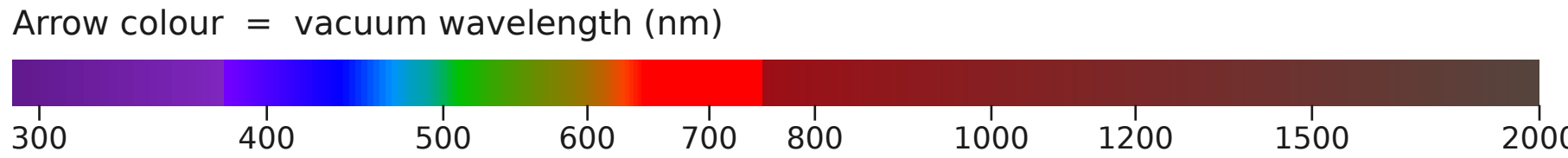


Thorium

Th (Th I). Strongest classified transitions below 2 μm (NIST): 67 levels, 71 lines.

Th⁺ ionisation limit 50867.00 cm⁻¹ = 6.30670 eV (196.591 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, = = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e r J' \rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
³ F ₂ - ² G ₃ ^o	372.0493	371.9435	805.7870	1.978	2.200e+07	-	-	first step of the Mainz three-step resonance ionization s...
³ F ₂ - ³ D ₃ ^o	380.4155	380.3075	788.0659	3.255	1.300e+08	-	-	alternative strong first step considered for resonance io...
³ F ₂ - (26113) ₂ ^o	382.9471	382.8384	782.8561	1.623	1.900e+07	-	-	first step of the KU Leuven in-gas-jet resonance ionizati...
³ G ₃ ^o - (38700) ₄	845.8742	845.6418	354.4173	-	-	-	-	second step of the Mainz resonance ionization scheme

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²²⁹Th hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
³ F ₂	39(9)	390(180)	F=1/2: +0.00 F=3/2: -117.00 F=5/2: -214.50 F=7/2: -146.25 F=9/2: +292.50
(26113) ₂ ^o	39(9)	-390(180)	F=1/2: -546.00 F=3/2: -312.00 F=5/2: -19.50 F=7/2: +185.25 F=9/2: +97.50

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
³ F ₂ - ³ G ₃ ^o	372.049	230-232	8994(1199)
³ F ₂ - ³ G ₃ ^o	372.049	228-232	18887(1199)
³ F ₂ - ³ G ₃ ^o	372.049	229-232	15589(1799)
³ F ₂ - (26113) ₂ ^o	382.947	229-232	19218(90)
³ G ₃ ^o - (38700) ₄	845.874	230-232	5096(1709)
³ G ₃ ^o - (38700) ₄	845.874	228-232	10193(1709)

Shift = ν(first isotope) - ν(second isotope).